

Strategic Allocation of Battery Energy Storage Systems in Multi-Service Markets: An EPEC Modeling Approach

Alexander Csekö¹, Daniel Horvath¹, Johann Auer¹, Sebastian
Zwickl-Bernhard¹

¹Energy Economics Group (EEG), Technische Universität Wien (TU Wien)

July 7, 2026



TECHNISCHE
UNIVERSITÄT
WIEN



Start and Current State

- 8 July 2026
- **Problem:** we need to validate and expand the model – some suspicion because of the fast convergence of the Gauss–Seidel algorithm
- **Problem:** the aFRR market looks very attractive

Investor	WACC	Arbitrage (€)	aFRR (€)
Investor 1	8.0%	-2,574.26	332,852.44
Investor 2	12.0%	4,521.00	292,986.68
Investor 3	15.0%	3,759.11	293,034.14
Investor 4	20.0%	3,532.13	293,041.04

Multi-service scenario: daily arbitrage vs. aFRR revenue.

- **Goal:** robust results from an extended model by end of September (Alex)

Resolution: Treat aFRR Activation as Uncertainty

- The old model rewards aFRR capacity, but does not fully price the **energy consequence** of activation.
- New plan: keep 24 hours and the same 5-node grid, but solve the LLP over activation scenarios $s \in \mathcal{S}$.
- Example scenarios: no activation, up-heavy, down-heavy, mixed, each with probability ω_s .
- The key correction is in the nodal energy balance, where the nodal price $\lambda_{n,t,s}$ emerges.

$$\sum_{g \in G_n} P_{g,t,s} + \sum_i \left(P_{i,n,t,s}^{discharge} - P_{i,n,t,s}^{charge} + P_{i,n,t,s}^{aFRR,up} - P_{i,n,t,s}^{aFRR,down} \right) + P_{n,t,s}^{shed} - D_{n,t}^{el} = NetInjection_{n,t,s}, \quad \forall n, t, s \quad (\lambda_{n,t,s})$$

- Upward activation injects energy into the node; downward activation absorbs energy from the node.
- Therefore, aFRR activation directly affects the LMP-forming balance instead of remaining a pure capacity payment.